

Vinmec × VinSmartFuture

Citation-grounded Medical Record Summarization PoC

Evidence-first AI draft workflow with clinician review, citation validation, auditability and proxy evaluation

Local staging PoC

Research/pilot foundation

Agenda

Cấu trúc theo style enterprise proposal: vấn đề trước, giải pháp sau, rồi kỹ thuật và lộ trình.

01 **Solution Introduction**

02 **Problem Assessment**

03 **Business Proposal**

04 **Technical Proposal**

05 **Delivery & Demo
Readiness**

06 **Research Pilot
Roadmap &
Governance**

Executive Summary

Một PoC có định hướng enterprise: evidence-first, review-first, audit-first.

Problem

Hồ sơ bệnh án phân mảnh; tóm tắt thủ công tốn thời gian; generic LLM dễ bỏ sót hoặc suy diễn.

Solution

RAG citation-grounded tạo AI draft theo patient/encounter scope, có citation validation và review workspace.

Evidence

Local staging PoC có UI doctor/admin, Flow 2.1 benchmark, artifacts, tests/build/runtime evidence.

Boundary

Chỉ dùng mock/de-identified/proxy data; clinician-review-only; không claim real EHR/clinical validation.

Thông điệp chính: PoC không cố thay bác sĩ; PoC làm bằng chứng, citation và rủi ro trở nên nhìn thấy được trước khi bác sĩ quyết định.

Why This Matters

Medical summarization không chỉ là rút gọn văn bản; đó là bài toán bằng chứng và trách nhiệm review.

Fragmented records

Thông tin nằm rải rác giữa note, encounter, medication, timeline và prior summaries.

LLM risk

Generic summarization có thể hallucinate, bỏ sót diagnosis/medication hoặc trộn ngữ cảnh sai bệnh nhân.

Clinical need

Bác sĩ cần nhìn được nguồn chứng cứ, claim nào unsupported và lịch sử ai đã duyệt/sửa.

Thiết kế đúng không phải “AI tự kết luận”, mà là một evidence-first workflow giúp bác sĩ kiểm tra bản nháp nhanh hơn và có dấu vết kiểm toán rõ hơn.



Current Challenge Assessment

Bảng vấn đề chuyển thành năng lực cần có trong một PoC enterprise-grade.

Challenge	Impact	Required capability
Hồ sơ nhiều nguồn	Bác sĩ mất thời gian tìm ngữ cảnh	Patient/encounter scoping + retrieval
Generic summary không có nguồn	Khó tin và khó review	Citation-first draft + evidence panel
Unsupported claim	Nguy cơ over-trust	Unsupported/conflict visibility
Benchmark chỉ ROUGE	Không đo grounding/safety proxy	Citation, omission, hallucination proxy
Demo thiếu audit	Khó bàn giao/kiểm tra	Review lifecycle + audit trail

Proposed Solution Overview

Một luồng nghiệp vụ ngắn: từ scope bệnh nhân đến draft có citation và audit.



Giá trị business của PoC: giảm ambiguity khi review, chuẩn hóa evidence visibility, và tạo nền để đo lường pilot nghiên cứu thay vì claim triển khai lâm sàng.

Product Entry — Role-based Workspace

Entry point cho doctor/admin demo với role-based navigation.

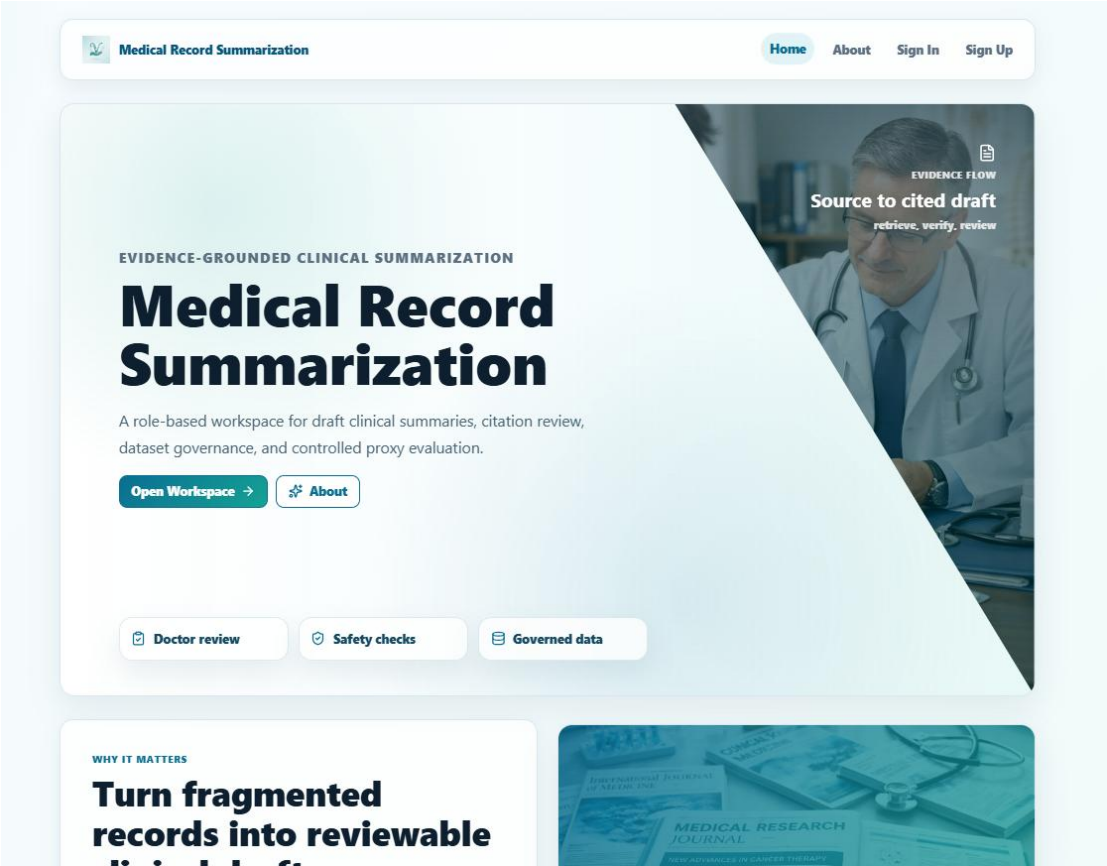


Figure 1 — Product landing page

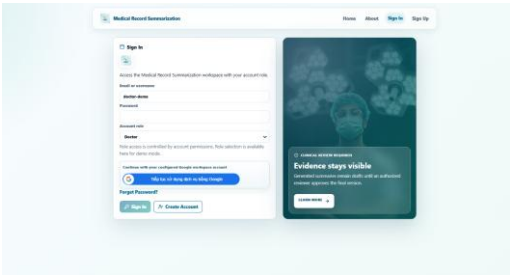


Figure 2 — Role-based login

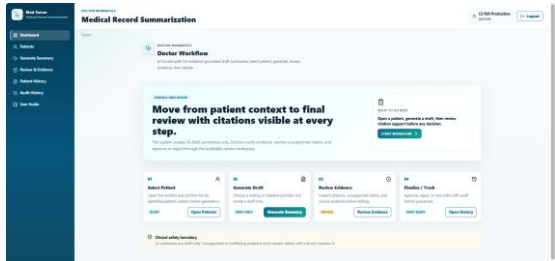


Figure 4 — Doctor workspace

Main message

- Workspace tách vai trò doctor/admin.
- Thông điệp safety boundary xuất hiện từ đầu.
- Clean text branding; không hàm ý phê duyệt chính thức.

Doctor Workflow — Patient Selection

Bác sĩ bắt đầu từ danh sách bệnh nhân de-identified và chọn đúng patient/encounter scope.

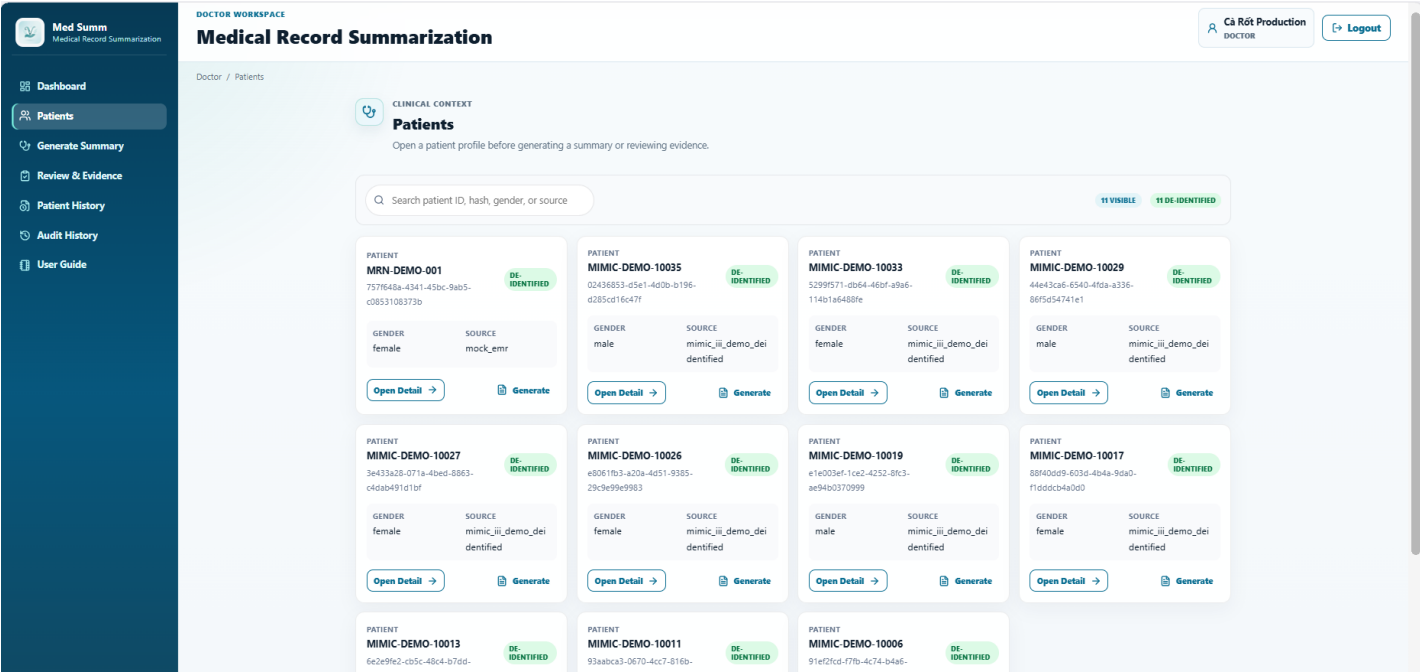


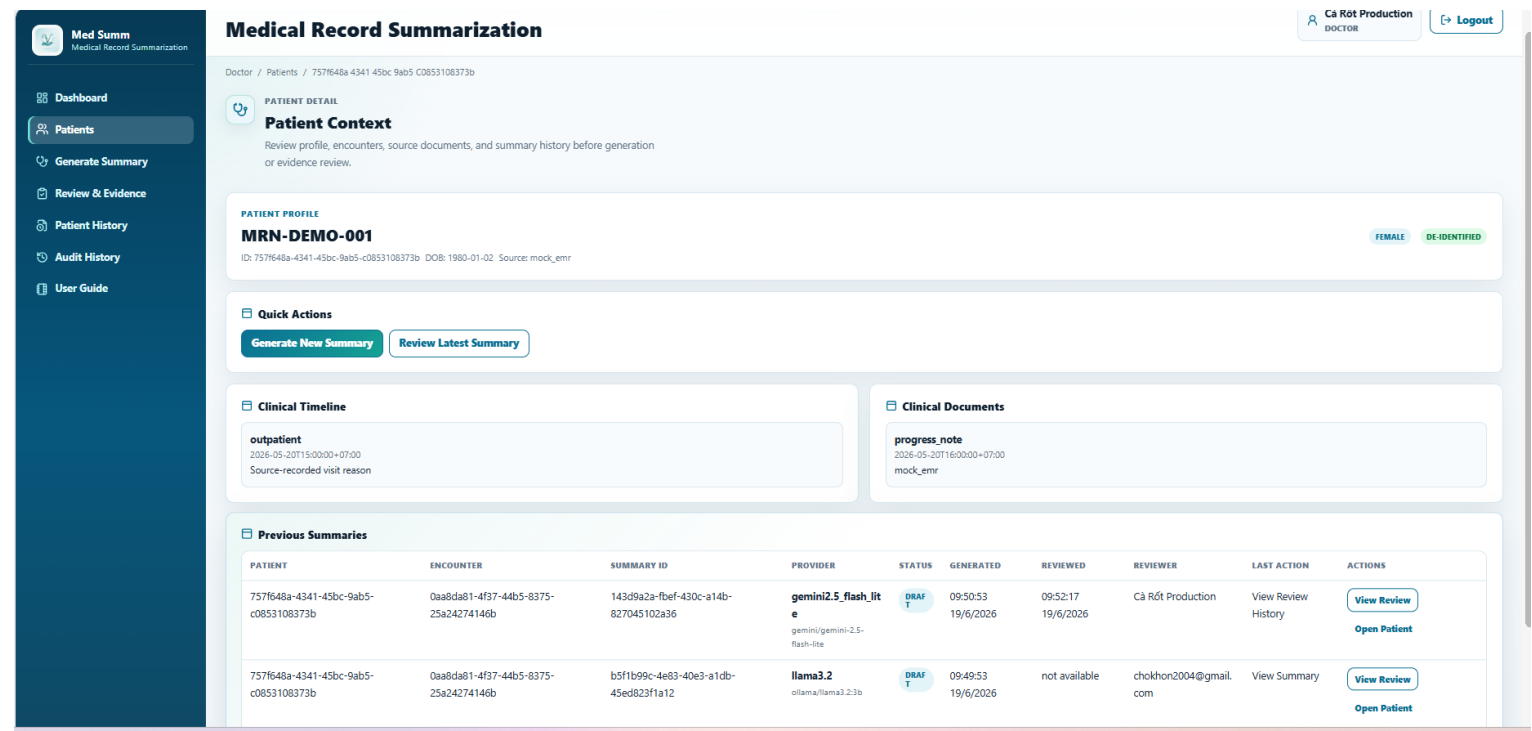
Figure 5 — De-identified patient list

What it proves

- Chỉ dùng mock/de-identified patient records.
- Search theo patient ID/hash/gender/source.
- Scope đúng bệnh nhân trước khi generate.
- Tránh trộn evidence sai bệnh nhân.

Doctor Workflow — Clinical Context

Patient context gom profile, timeline, source documents và previous summaries.



Clinical context layer

- Profile + encounter timeline.
- Source document list.
- Previous summary history.
- Context được xem trước generation/review.

Figure 6 — Patient context and timeline

Doctor Workflow — Generate AI Draft

RAG evidence-first: retrieve trước, generate sau, draft luôn cần review.

Med Summ
Medical Record Summarization

Dashboard

Patients

Generate Summary

Review & Evidence

Patient History

Audit History

User Guide

DOCTOR GOLDEN PATH
Generate Summary

Create a draft patient snapshot. Evidence review, editing, and approval happen on the Review & Evidence page.

FLOW 2 / RAG EVIDENCE-FIRST

Doctor generation now uses patient-scoped MiniLM + Qdrant retrieval before provider inference.
Notes are chunked, embedded, retrieved by clinical section, checked by a quality gate, then sent to the selected provider as cited context.

Patient & Source

Patient search

Search patient ID, hash, or gender

MRN-DEMO-001

female

DE-ID

MIMIC-DEMO-10035

male

DE-ID

MIMIC-DEMO-10033

female

DE-ID

MIMIC-DEMO-10029

male

DE-ID

MIMIC-DEMO-10027

female

DE-ID

MIMIC-DEMO-10026

female

DE-ID

MIMIC-DEMO-10019

male

DE-ID

MIMIC-DEMO-10017

female

DE-ID

MIMIC-DEMO-10013

female

DE-ID

MIMIC-DEMO-10011

female

DE-ID

Selected patient:MRN-DEMO-001

DE-IDENTIFIED

Encounter

outpatient

Source document

progress_note

Provider

Summary Provider

SELECTED Gemini 2.5 Flash Lite

READY

Change

Patient

MRN-DEMO-001

Encounter

outpatient

Type

API provider

Context mode

RAG MINILM + QDRANT

Draft only. Review evidence before approval.

PROVIDER READY

WORKER READY

Draft generated

125

JOB 34DBE3DD...D208

DRAFT_READY

Retrieval quality gate: warnings: Missing required: none.

Patient scope

MiniLM + Qdrant section-aware evidence search

Retrieve evidence

MiniLM + Qdrant section-aware evidence search

Build clinical context

Diagnosis, medications, timeline, diagnostics, assessment, plan

Provider generation

Gemini 2.5 Flash Lite cloud provider

Citation validation

Checking unsupported claims and missing evidence

Draft ready

Open Review & Evidence before approval

Generate Draft

Draft Preview

DRAFT

Summary ID

16f5032a...bf6395

Provider

gemini2.5_flash_lite

Generated

16c5439 24/6/2026

Patient Snapshot

- Gender: female [chunk:7f48bc24-899c-5690-a3b1-7e11fe592cba]
- Age: 46 [chunk:7f48bc24-899c-5690-a3b1-7e11fe592cba]
- Data status: de-identified [chunk:7f48bc24-899c-5690-a3b1-7e11fe592cba]

Diagnosis Evidence

- Condition: Source-recorded condition [chunk:e2c38bc8-a775-5d04-bdc6-3d62f489aa5]
- Clinical status: active [chunk:e2c38bc8-a775-5d04-bdc6-3d62f489aa5]

Medication Evidence

- Medication: Source-recorded medication [chunk:27174b40-52d8-5a69-80ce-5efdbca2bca9]
- Status: active [chunk:27174b40-52d8-5a69-80ce-5efdbca2bca9]
- Dosage: source-recorded dosage [chunk:27174b40-52d8-5a69-80ce-5efdbca2bca9]

Timeline Evidence

- Encounter type: outpatient [chunk:7cad3932-cf99-5]

Needs Clinician Review

- Provider output requires clinician review before clinical use.

Review Evidence

1. Scope

Patient + encounter + source document

2. Retrieve

MiniLM + Qdrant / evidence search

3. Draft

Provider tạo AI-generated draft

Draft only — bác sĩ review evidence trước khi approve/reject.

Figure 7 — RAG evidence-first generate summary

Ranh giới: dữ liệu mock/de-identified/proxy; AI chỉ tạo bản nháp; bác sĩ duyệt; không chẩn đoán/tư vấn điều trị/kê đơn; không phải triển khai bệnh viện hay xác nhận an toàn/hiệu quả lâm sàng.

Evidence Quality Gate

Gate làm rủi ro hiển thị trước khi bác sĩ quyết định.

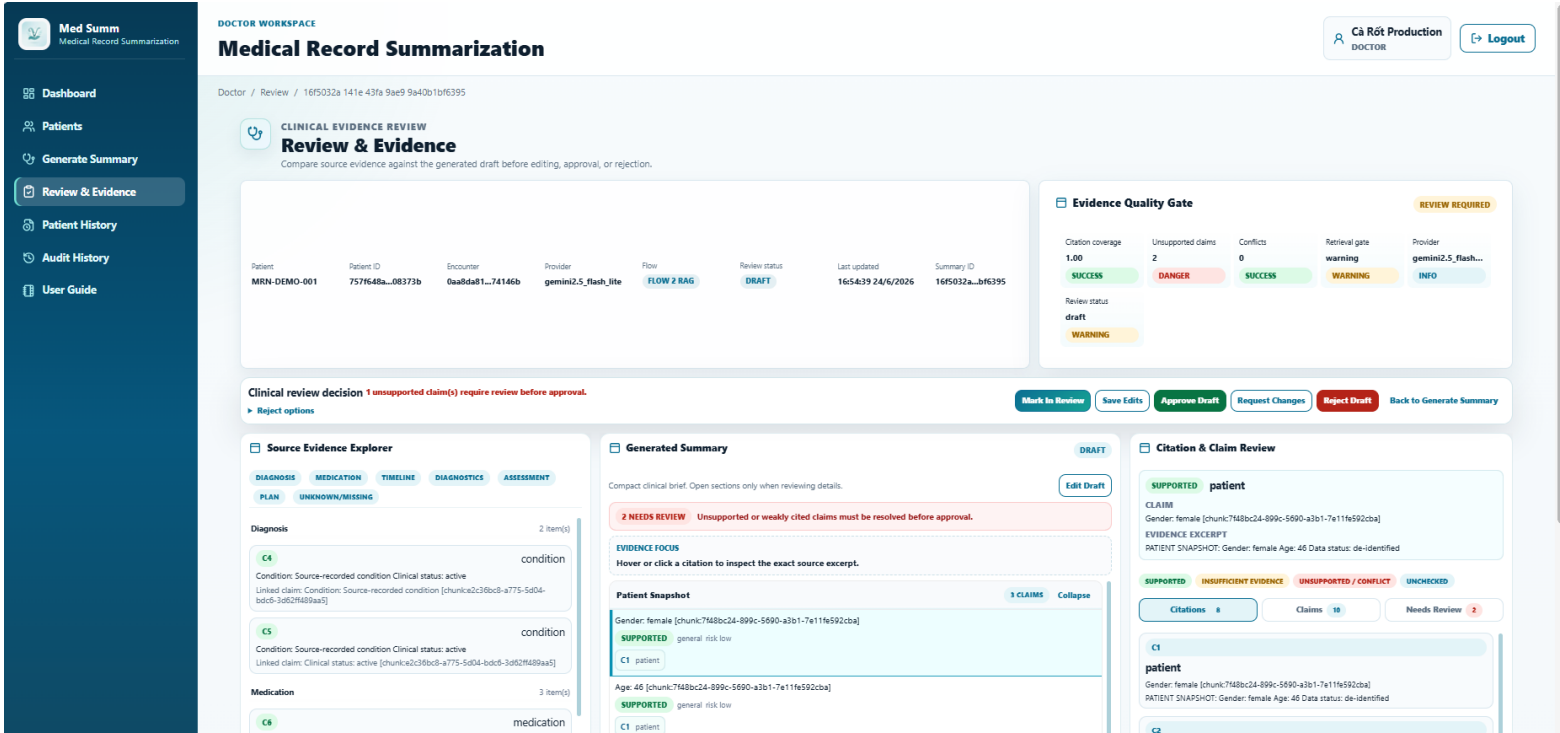


Figure 8 — Review and evidence quality gate

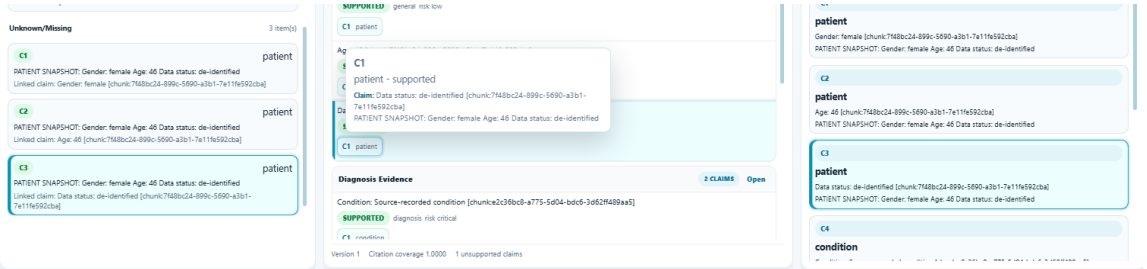
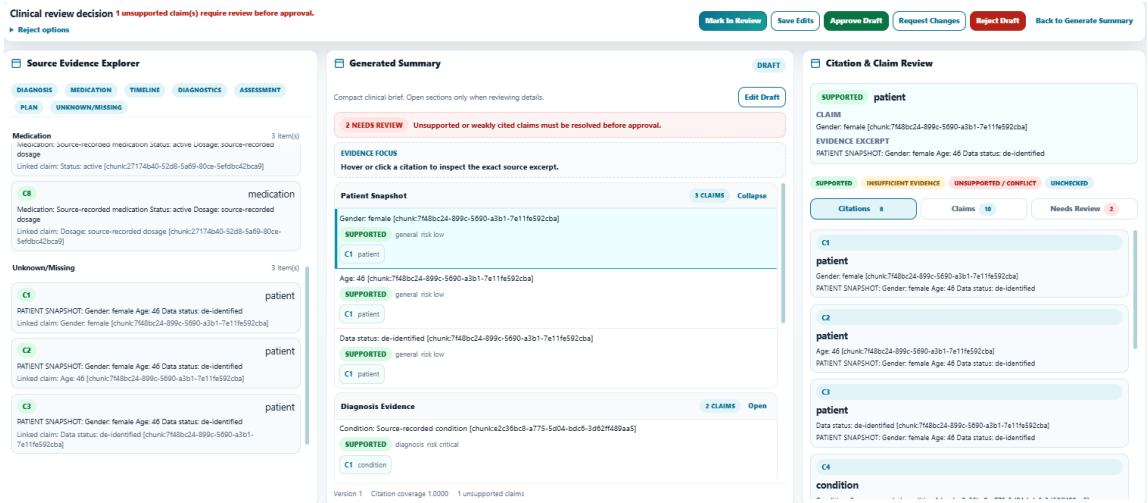
Quality signals

- Citation coverage.
- Unsupported claims.
- Conflicts.
- Retrieval warning.
- Review status.

Không ép generation khi evidence yếu;
cảnh báo phải visible.

Citation-first Review

Mỗi claim quan trọng cần trace được về evidence source.



Main message: citation không chỉ để trang trí; citation là cơ chế review, debug và giảm unsupported claim.

Human-in-the-loop Decision

Bác sĩ là người quyết định cuối; AI chỉ tạo draft có thể sửa hoặc từ chối.

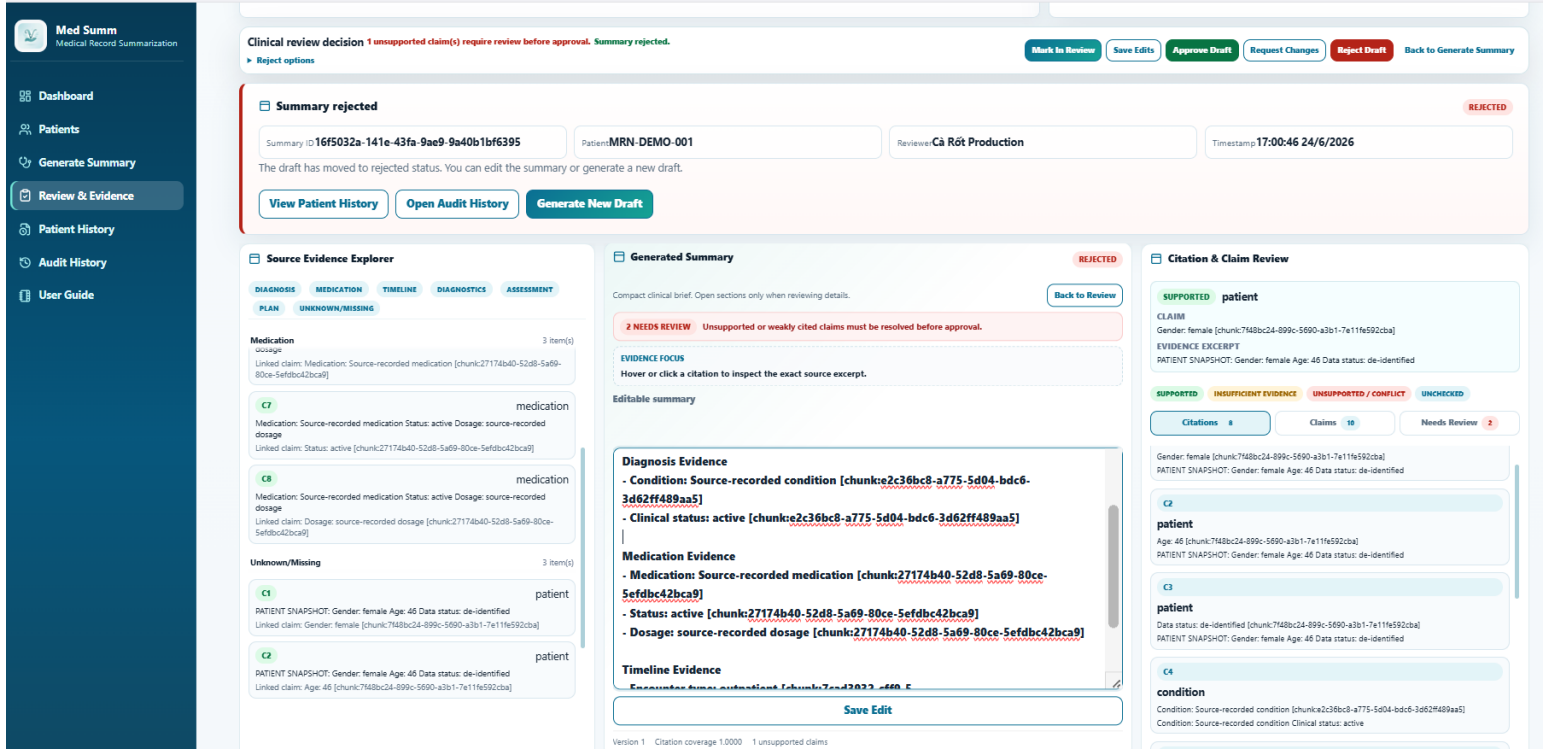


Figure 11 — Editable draft and reject decision

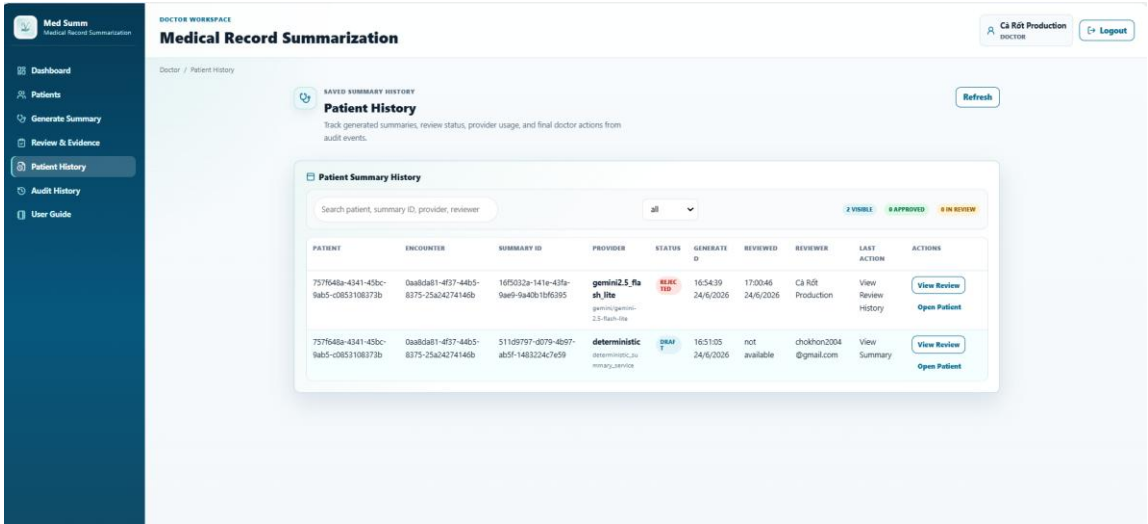
Clinician authority

- Edit draft.
- Approve draft.
- Request changes.
- Reject draft.
- Review decision captured.

Không có autonomous diagnosis, treatment recommendation hoặc prescribing.

Summary Lifecycle and Audit

Lifecycle + audit trail giúp reviewer thấy ai làm gì, khi nào, trên summary nào.



							Each file
165542 24/6/2026	757f648a-4341-43bc-9ab5-c0853108373b	not applicable	not applicable	View Documents	not applicable	RECORD ID	chokhon2004@gmail.com
165542 24/6/2026	757f648a-4341-43bc-9ab5-c0853108373b	not applicable	not applicable	View Patient	not applicable	RECORD ID	chokhon2004@gmail.com
165542 24/6/2026	757f648a-4341-43bc-9ab5-c0853108373b	not applicable	not applicable	View Encounters	not applicable	RECORD ID	chokhon2004@gmail.com
165542 24/6/2026	757f648a-4341-43bc-9ab5-c0853108373b	0aadb8a1-4f37-44b5-8375-25a24274148b	16f5032a-141e-43fa-9ae9-9a40b1b6395	View Review History	gemini2.5_flash_lite	DRAFT	CA R&B Production
165542 24/6/2026	757f648a-4341-43bc-9ab5-c0853108373b	0aadb8a1-4f37-44b5-8375-25a24274148b	16f5032a-141e-43fa-9ae9-9a40b1b6395	View Summary	gemini2.5_flash_lite	DRAFT	chokhon2004@gmail.com
165542 24/6/2026	757f648a-4341-43bc-9ab5-c0853108373b	0aadb8a1-4f37-44b5-8375-25a24274148b	16f5032a-141e-43fa-9ae9-9a40b1b6395	View Summary	gemini2.5_flash_lite	DRAFT	chokhon2004@gmail.com
165441 24/6/2026	757f648a-4341-43bc-9ab5-c0853108373b	0aadb8a1-4f37-44b5-8375-25a24274148b	16f5032a-141e-43fa-9ae9-9a40b1b6395	View Summary	gemini2.5_flash_lite	DRAFT	chokhon2004@gmail.com
165439 24/6/2026	757f648a-4341-43bc-9ab5-c0853108373b	0aadb8a1-4f37-44b5-8375-25a24274148b	16f5032a-141e-43fa-9ae9-9a40b1b6395	Generate Summary	gemini2.5_flash_lite	DRAFT	chokhon2004@gmail.com

Auditability là yêu cầu nền cho nghiên cứu/pilot: không chỉ output tốt, mà phải trace được hành động và trạng thái review.



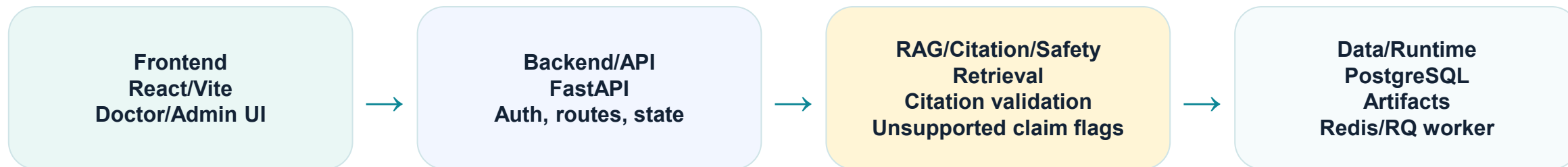
User / Role / Function Mapping

Map nhu cầu người dùng sang feature và giá trị đề xuất.

User	Pain point	Feature	Value
Doctor	Mất thời gian đọc record dài	Citation-grounded draft + review	Review nhanh hơn, evidence visible
Admin	Khó nhìn benchmark/provider	Admin evaluation dashboards	Theo dõi readiness và artifacts
Technical Reviewer	Cần kiểm chứng runtime	Docker/test/build evidence	Demo readiness rõ ràng
Research Evaluator	Cần proxy metrics sâu	ROUGE, BERTScore, grounding, omission	Đánh giá có chiều nghiên cứu
Mentor/Reviewer	Cần câu chuyện end-to-end	Proposal deck + evidence package	Dễ review nhanh và phản biện

Technical Architecture

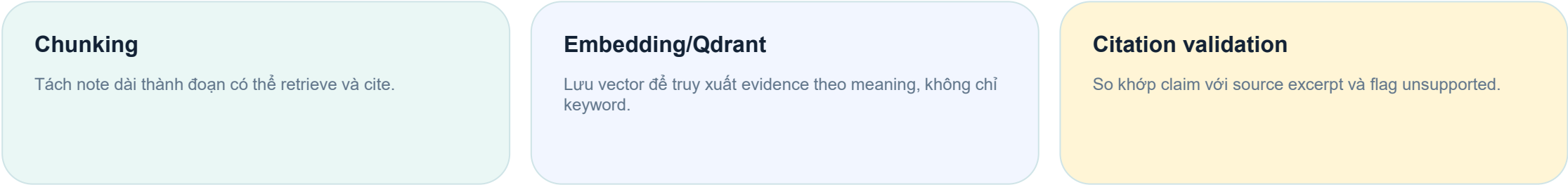
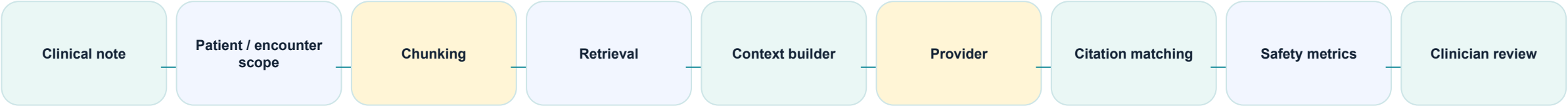
Kiến trúc staging demo: frontend, FastAPI, services, storage/artifacts và worker.



Boundary: local staging architecture prepared for demo evidence; public cloud deployment and real EHR integration are future/governed work.

RAG and Citation Pipeline

RAG trong bài toán này là hạ tầng bằng chứng, không chỉ là cách gọi LLM.



Evaluation Framework

Proxy evaluation phải đo cả chất lượng summary lẫn evidence grounding.

ROUGE-L

Similarity proxy

BERTScore

Semantic similarity

Citation coverage

Evidence coverage

Unsupported claim rate

Safety proxy

Factuality proxy

Grounding quality

Timeline completeness

Clinical structure

Hallucination proxy

Unsupported content

Critical omission proxy

Missing key facts

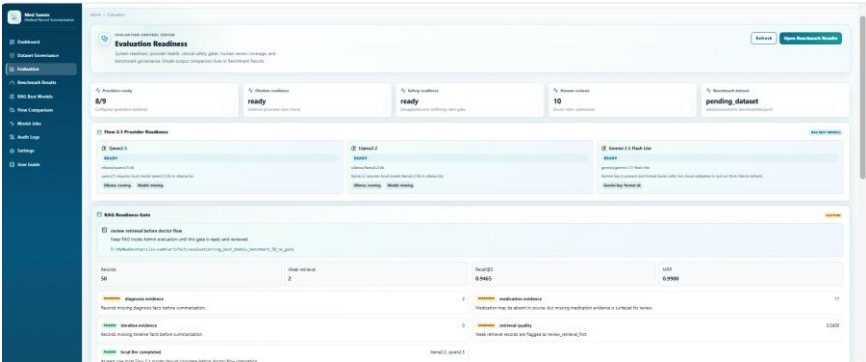


Figure 14 — Admin evaluation readiness

Không chỉ ROUGE/BERTScore: citation và unsupported claim là phần bắt buộc trong medical summarization proxy evaluation.

Benchmark Result Summary

Flow 2.1 no-gate: 50 records × 5 providers = 250/250 predictions.

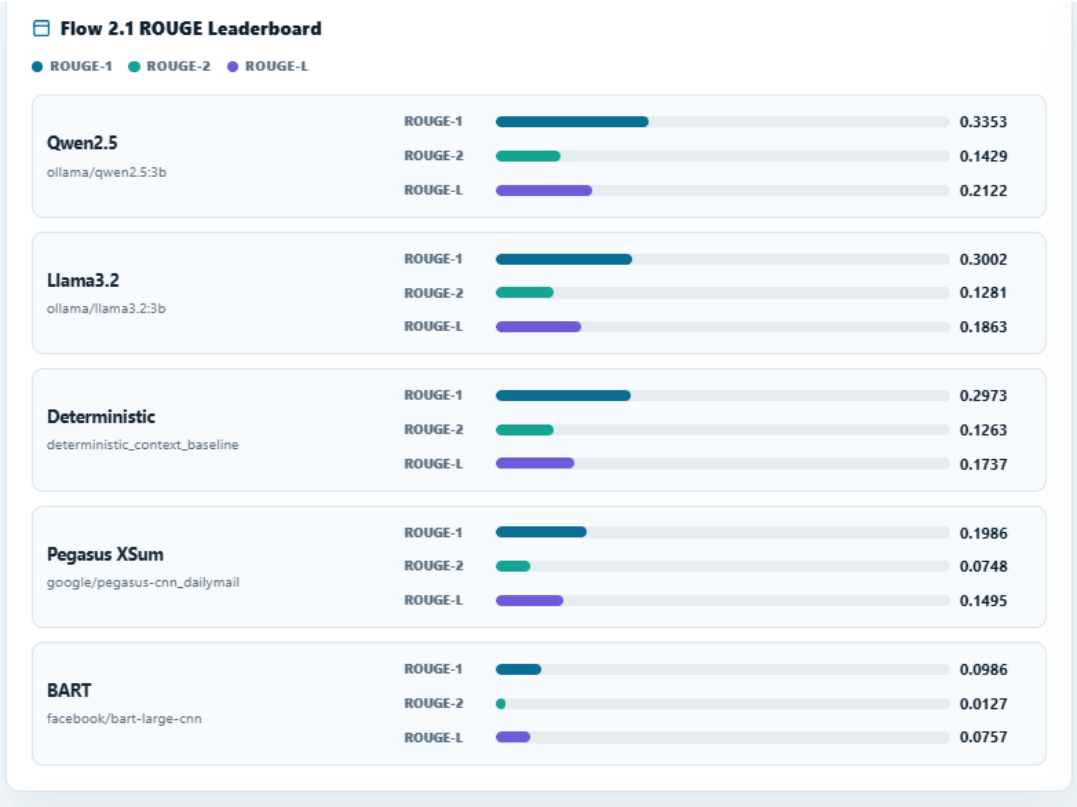


Figure 16 — Provider ROUGE leaderboard

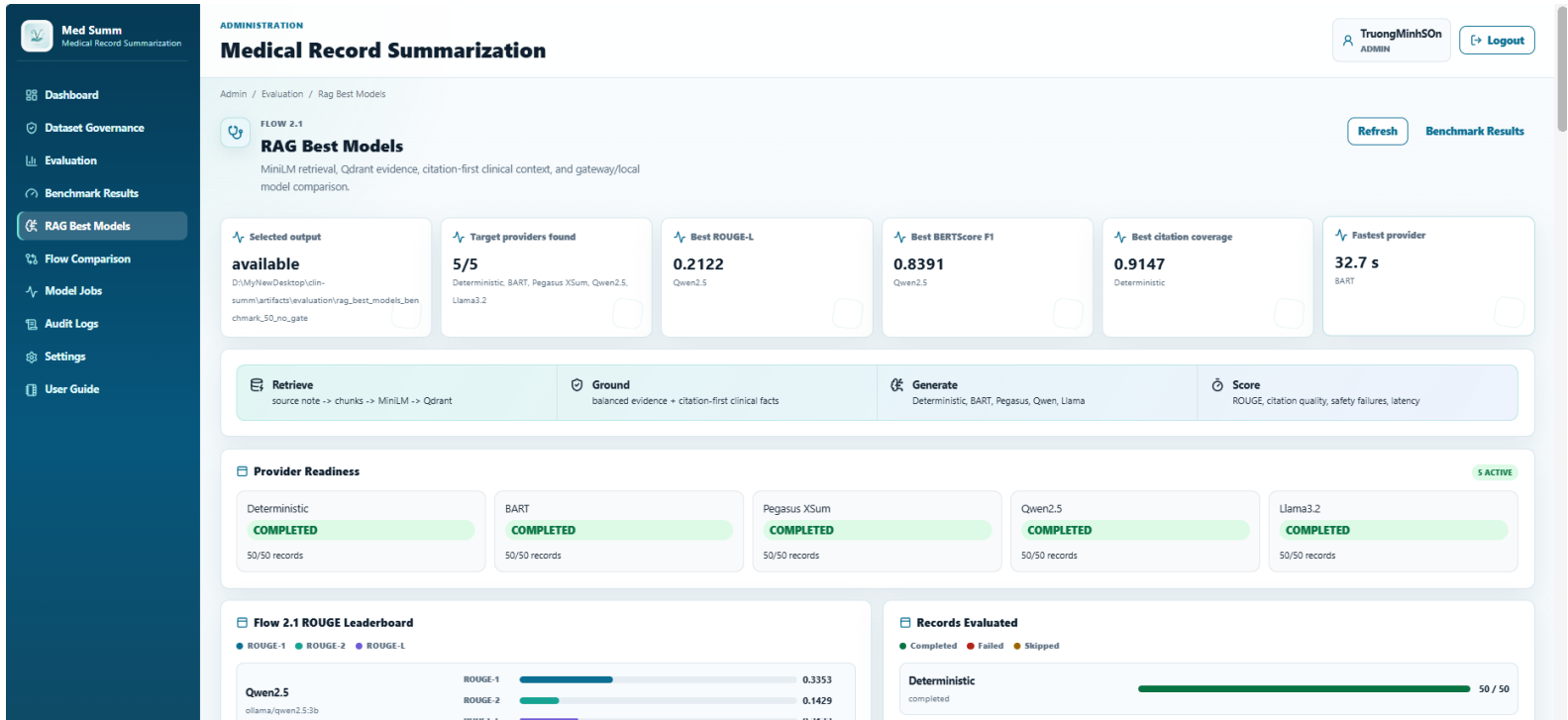
Qwen2.5 metric		Value
ROUGE-L		0.2122
BERTScore F1		0.8391
Citation coverage		0.8884
Factuality proxy		0.8713
Critical omission		0.4460

Interpretation

- Qwen2.5 là strongest generative PoC provider trong proxy run.
- Deterministic là smoke/control provider đáng tin cậy nhất.
- Kết quả là proxy evaluation, không phải clinical validation.

Admin Evaluation Dashboard

Admin nhìn được Flow 2.1 benchmark visibility và provider completion.



Admin value

- 5/5 target providers found.
- Provider completion visible.
- ROUGE/BERTScore/citation summary visible.
- Artifacts path visible for reproducibility.

Figure 15 — RAG best models admin overview

Grounding Metrics and Error Analysis

Không chỉ biết model nào thắng; reviewer thấy vì sao model fail.

Evidence Grounding				
Best faithfulness proxy				
PER_RECORD_METRICS.CSV				
Deterministic				
Deterministic	Citation coverage	Unsupported claim rate	Faithfulness proxy	Critical omission
deterministic_context_baseline	0.9147	0.0000	0.9071	0.3688
	Timeline completeness			
	0.6667			
BART				
BART	Citation coverage	Unsupported claim rate	Faithfulness proxy	Critical omission
facebook/bart-large-cnn	0.1307	0.0000	0.7585	0.9583
	Timeline completeness			
	0.0645			
Pegasus XSum				
Pegasus XSum	Citation coverage	Unsupported claim rate	Faithfulness proxy	Critical omission
google/pegasus-cnn_dailymail	0.3800	0.0100	0.7626	0.9236
	Timeline completeness			
	0.1290			
Qwen2.5				
Qwen2.5	Citation coverage	Unsupported claim rate	Faithfulness proxy	Critical omission
ollama/qwen2.5:3b	0.8884	0.0346	0.8713	0.4460
	Timeline completeness			
	0.4785			
Llama3.2				
Llama3.2	Citation coverage	Unsupported claim rate	Faithfulness proxy	Critical omission
ollama/llama3.2:3b	0.8620	0.0618	0.8413	0.5108
	Timeline completeness			
	0.4570			

Per-Record Failure Analysis

All models

All failure types

Export case

Total records

50

Filtered records

50

Selected model

All

Selected failure

All

hallucinated content 46

incomplete summary 42

missing diagnosis 21

missing medication 5

missing timeline 31

retrieval-related failure 36

source data limitation 31

no major proxy failure detected 1

multiclinsum_is_en_10041

hallucinated content, incomplete summary, missing diagnosis

Deterministic / BART / Pegasus XSum / Qwen2.5 / llama2

multiclinsum_is_en_10024

multiclinsum

hallucinated content, incomplete summary

multiclinsum_is_en_1002

incomplete summary, missing timeline, retrieval-related failure

Deterministic / BART / Pegasus XSum / Qwen2.5 / llama2

multiclinsum_is_en_10005

hallucinated content, incomplete summary, missing timeline

Deterministic / BART / Pegasus XSum / Qwen2.5 / llama2

multiclinsum_is_en_10033

hallucinated content, incomplete summary, missing diagnosis

Deterministic / BART / Pegasus XSum / Qwen2.5 / llama2

multiclinsum_is_en_1000

hallucinated content, incomplete summary, missing timeline

Deterministic / BART / Pegasus XSum / Qwen2.5 / llama2

multiclinsum_is_en_10024

multiclinsum

Input note

A 34-year-old male patient was found lying unconscious with a head injury on the road after participating in a half marathon in the spring. It was a sunny day with a maximum temperature of 34.2 °C and a humidity of 54%. A physician who was transported by helicopter to check on the patient reported that his Glasgow Coma Scale score was 6 and that he presented marked restlessness. His blood pressure was 110/80 mmHg. His heart rate was 140 beats per minute (BPM), his respiratory rate was 40 breaths per minute (BPM), and his axillary temperature was 40.8 °C. He was transported to our hospital by a *ground ambulance* after the initiation of cardiac arrest and the

Reference summary

(L)AK level increased to 0.832 IU/L on the 11th day of hospitalization and then showed a tendency to transiently decrease. The patient was evaluated on the eighth day of hospitalization after the improvement of his laboratory data. From the ninth day of hospitalization, gradual rehabilitation was initiated. However, he felt pain in both legs and his CPK level increased again. Despite the cessation of all drugs and rehabilitation, his CPK level increased to 105,945 IU/L on the 15th day of hospitalization. Fortunately, his CPK level decreased with the fluid management. The patient's rehabilitation was restarted after his CPK level fell to <1000 IU/L. On the 31st day of hospitalization, his CK level

Retrieved evidence

You are reading a strict clinical RAG context. Use the FACTS blocks first and the retrieved evidence only as support. Every explicit claim must be supported by an explicit chunk id. Do not borrow facts across sections. Do not complete truncated source text. Important: "not present in retrieved evidence" means unknown, not absent.

[CITATION_FIRST_CLINICAL_FACTS]

[RETRIEVED_FACTS]

[FACTS]

Deterministic

deterministic_context_baseline

COMPLETED

ROUGE-L

Faithfulness

Unsupported

0.0106

1.0000

0.0000

Diagnosis miss

Medication miss

Timeline

0.0000

n/a

1.0000

NO MAJOR PROXY FAILURE DETECTED

[54633b28-2497-561c-b6b2-5cc20b9a0333] His general condition was initially thought to be improving; however, his smoldering rhabdomyolysis suddenly flared up with a marked increase in his CPK level (105,231 IU/L on the 18th day of hospitalization) when the steroid

BART

facebook/bart-large-cnn

COMPLETED

ROUGE-L

Faithfulness

Unsupported

0.0330

0.6500

0.0000

Diagnosis miss

Medication miss

Timeline

1.0000

n/a

0.0000

INCOMPLETE SUMMARY

MISSING DIAGNOSIS

MISSING TIMELINE

You are reading a strict clinical RAG context. Use the FACTS blocks first and the retrieved evidence only as support. Every clinical claim must be supported by an explicit chunk id. Do not borrow facts across sections. "not present in retrieved

Pegasus

XSum/google/pegasus-cnn_dailymail

COMPLETED

ROUGE-L

Faithfulness

Unsupported

0.0709

0.6500

0.0000

Diagnosis miss

Medication miss

Timeline

1.0000

n/a

0.0000

INCOMPLETE SUMMARY

MISSING DIAGNOSIS

MISSING TIMELINE

His general condition was initially thought to be improving. <n>However, his smoldering rhabdomyolysis suddenly flared up with a marked increase in his

Qwen2.5

llama/qwen2.5:3b

COMPLETED

ROUGE-L

Faithfulness

Unsupported

0.2096

0.8611

0.1111

Diagnosis miss

Medication miss

Timeline

0.5000

n/a

1.0000

HALLUCINATED CONTENT

INCOMPLETE SUMMARY

MISSING DIAGNOSIS

His general condition initially improved, but he experienced a sudden flare-up in rhabdomyolysis with an increase in his CPK level (105,231 IU/L) on the 18th day of hospitalization when steroid dosage was reduced and rehabilitation was initiated. (See previous fresh frozen

Llama3.2

llama/llama3.2:3b

COMPLETED

ROUGE-L

Faithfulness

Unsupported

0.2843

0.9983

0.0000

Diagnosis miss

Medication miss

Timeline

0.0000

n/a

1.0000

NO MAJOR PROXY FAILURE DETECTED

""Clinical Summary"" ""Diagnosis"" Acute Hepatic Failure ""Retrieved Evidence"" Medication information was not present in retrieved evidence. ""Timeline"" Day 2: Received fresh frozen plasma and a platelet transfusion due to elevated AST, ALT, prothrombin

Error analysis giúp chuyển benchmark thành research discussion: omission, hallucination proxy, retrieval failure, source limitation.

RAG vs Raw Context Comparison

Raw có thể tốt với note ngắn; RAG hữu ích hơn khi cần evidence-first, multi-document và citation review.

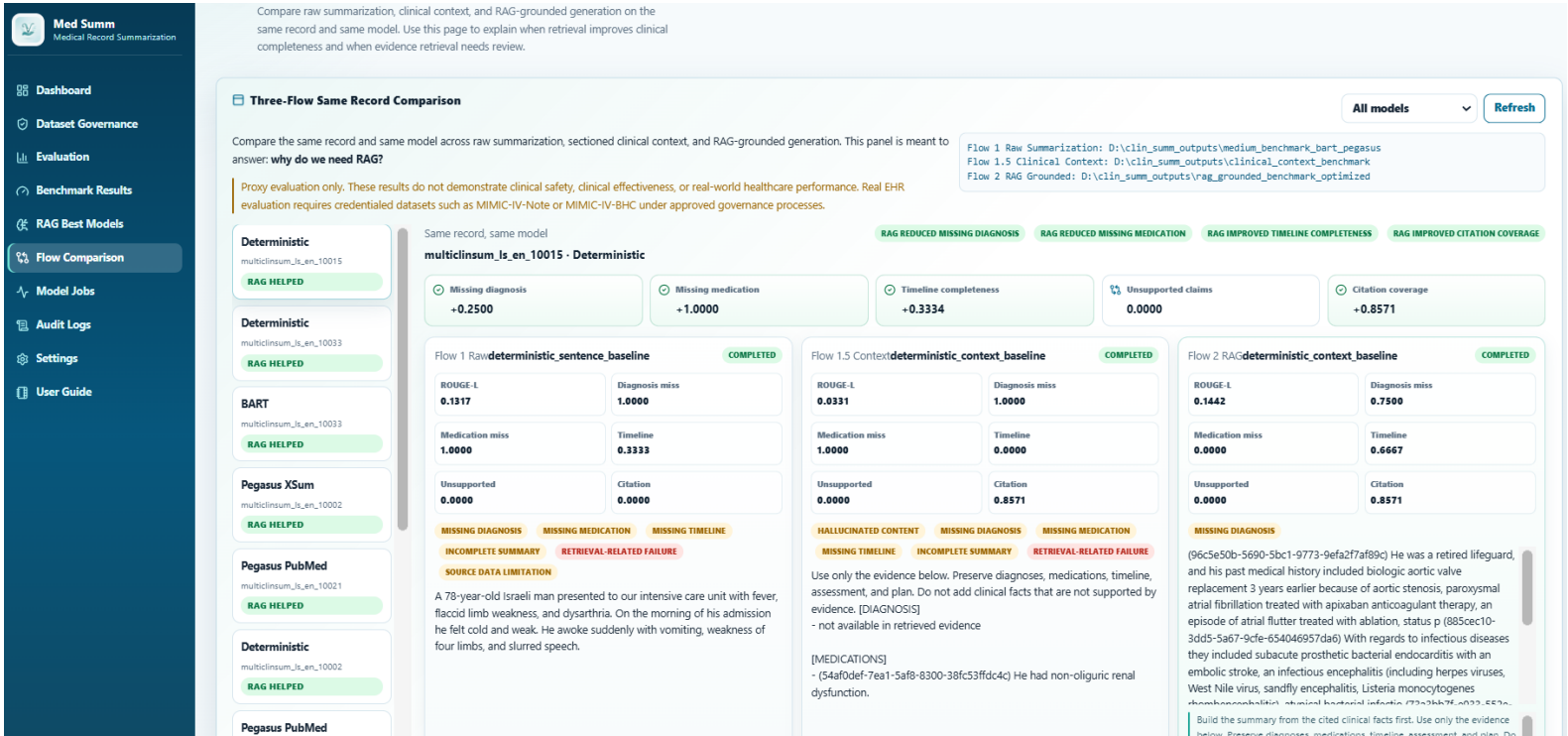


Figure 18 — RAG vs raw/context comparison

Balanced interpretation

- RAG là evidence infrastructure.
- Raw context vẫn là baseline hữu ích.
- Adaptive strategy nên được nghiên cứu ở pilot.
- Không claim RAG luôn tốt hơn mọi trường hợp.

Reproducibility and Artifacts

Benchmark không chỉ là screenshot; artifacts giúp reviewer kiểm tra lại.

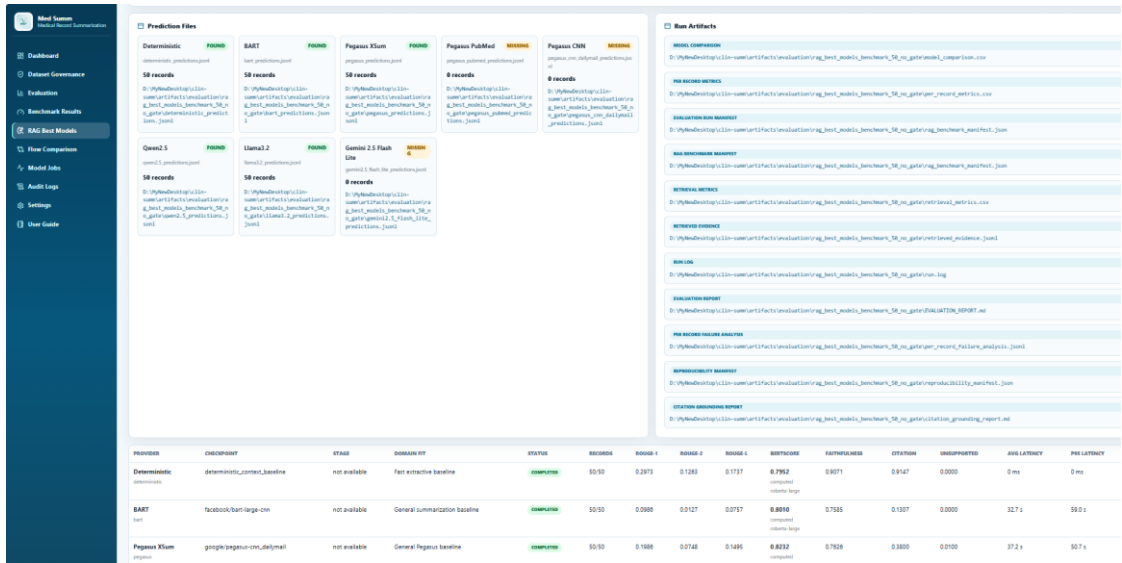


Figure 20 — RAG benchmark artifacts and run files

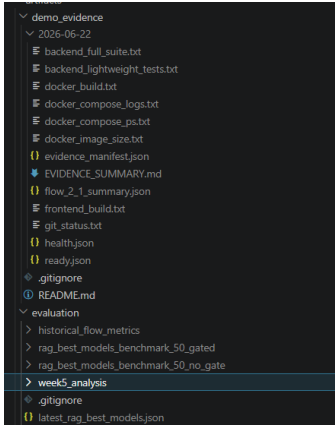


Figure 22 — Evidence package folder

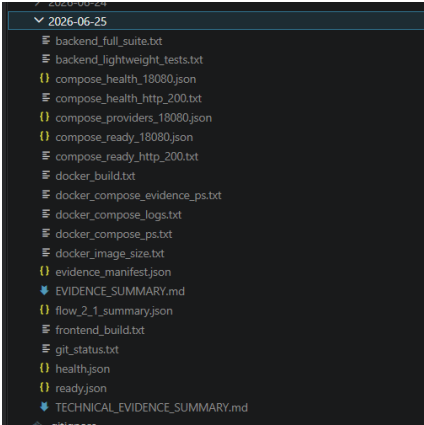


Figure 23 — Latest running evidence

Artifact package

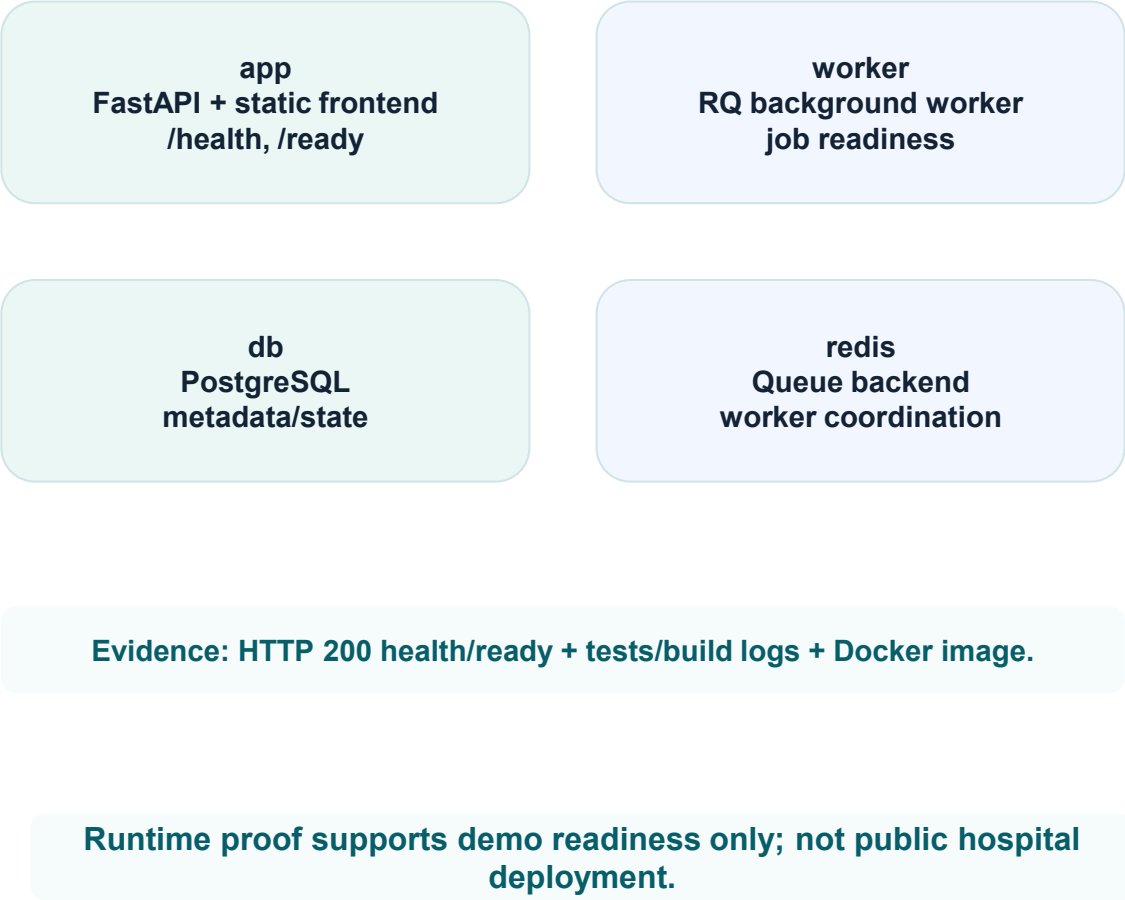
- Prediction JSONL, metrics CSV, manifests.
- Health/readiness responses.
- Backend/frontend/docker logs.
- Evidence package supports repeatability.

Demo Staging Architecture

Docker Compose local staging path validates app, worker, PostgreSQL, Redis, /health và /ready.

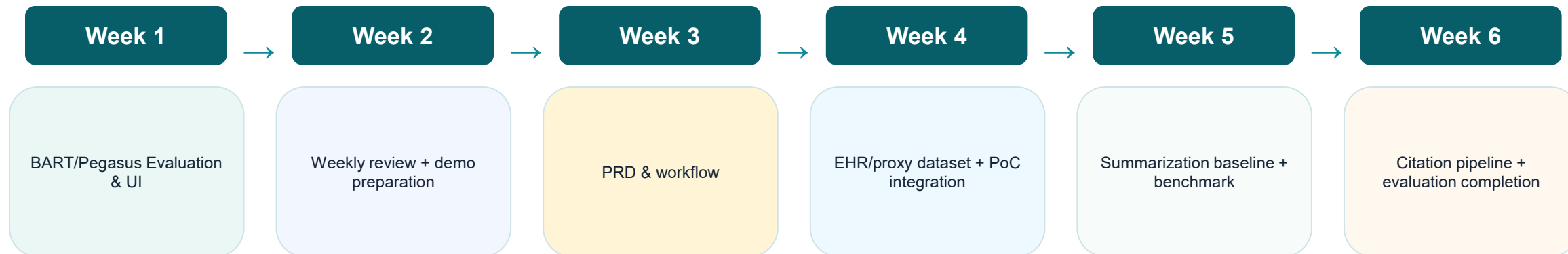
Bảng chứng kỹ thuật	Command terminal cần chụp	Kết quả cần thấy
Docker Compose services running	<code>docker compose ps</code>	<code>app , db , redis</code> running/healthy
Health endpoint reachable	<code>Invoke-WebRequest http://127.0.0.1:8080/health Select-Object StatusCode,Content</code>	<code>StatusCode: 200</code>
Readiness endpoint reachable	<code>Invoke-WebRequest http://127.0.0.1:8080/ready Select-Object StatusCode,Content</code>	<code>StatusCode: 200</code>
Backend tests pass	<code>python -m pytest backend\tests -p no:cacheprovider -q</code>	<code>passed</code>
Frontend build pass	<code>cd frontend rồi npm run build</code>	<code>built in ...</code> hoặc build success
Docker build pass	<code>docker build -t clin-summ- local:demo .</code>	<code>Successfully</code> <code>built /image</code> created

Figure 21 — Technical system running checklist



Delivery Schedule

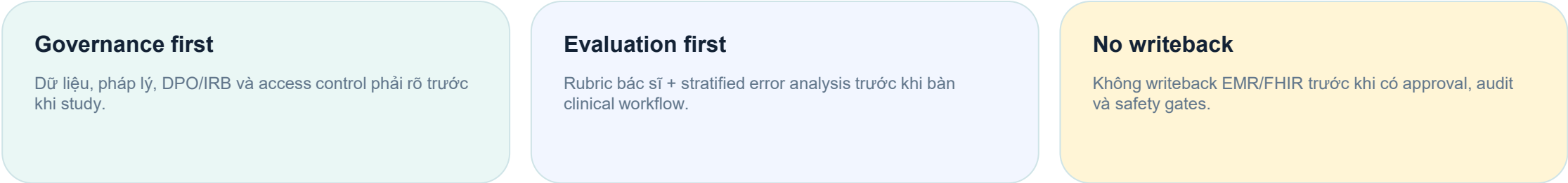
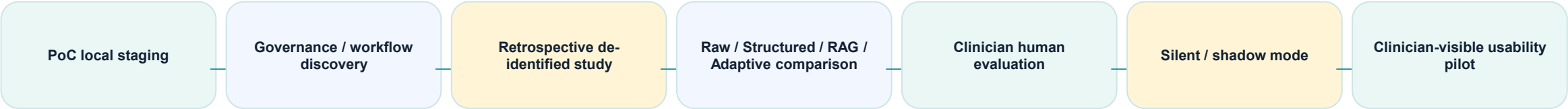
Timeline Week 1–Week 6 theo artifact/delivery hiện tại.



Delivery narrative: UI + workflow + proxy data + RAG benchmark + citation/evaluation completion.

Vinmec Research Pilot Appendix

Bước hợp lý là research-first pilot, không phải rollout hospital workflow.





Limitations and Risk Controls

Risk table cho proposal enterprise: mitigation rõ, limitation còn lại rõ.

Risk	Mitigation	Remaining limitation
Wrong-patient evidence	Patient/encounter scope + audit	Cần kiểm thử real workflow
Unsupported diagnosis	Unsupported/conflict flag	Proxy metric chưa thay bác sĩ
Medication/allergy error	Citation review + critical omission proxy	Cần human clinical rubric
PHI leakage	Mock/de-identified data + private-by-default	Cần governance trước real data
Over-trust	Draft-only UI + reviewer action	Cần training và policy
Citation mismatch	Citation matching + evidence panel	Cần manual validation sample

Roadmap and Conclusion

Kết luận bảo thủ: tăng evidence, human review và governance trước khi mở rộng.

1. Final demo evidence

Record demo + package logs/screenshots/artifacts.

2. Human evaluation

Rubric bác sĩ, blinded samples, disagreement review.

3. Data diversity

Stratified error analysis theo note type, specialty, length.

4. Adaptive routing

So sánh Raw / Structured / RAG / Adaptive.

5. Optional public demo

Chỉ khi resources/governance cho phép.

6. Real EHR study

Chỉ dưới approved governance, de-identification và audit.

Final positioning: local staging demo-ready PoC + production-readiness roadmap + research/pilot foundation.

Discussion focus

Scope

PoC evidence-first, not clinical deployment.

Evaluation

Proxy metrics + clinician rubric next.

Governance

Real data only under approved process.

Thank you — cảm ơn thầy/cô và mentor đã review.